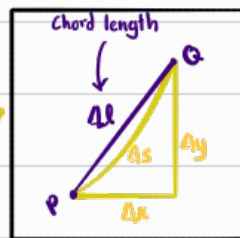
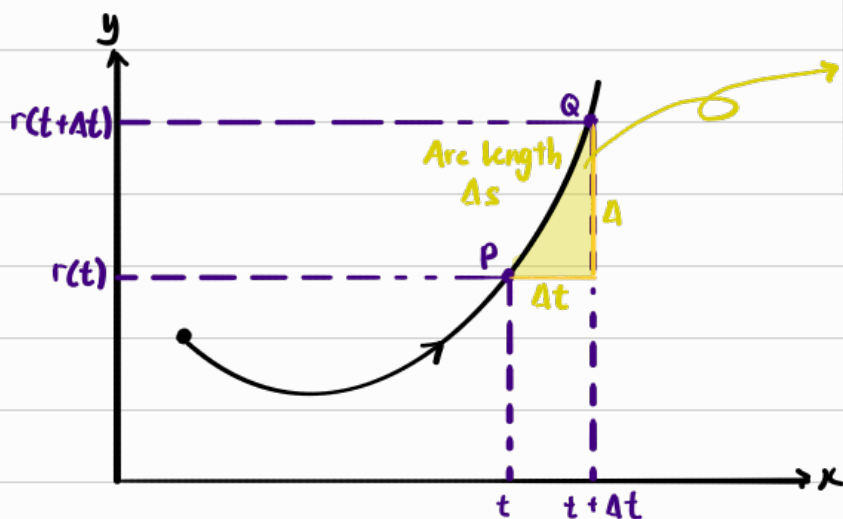


Arc length, 2D

Assume we have a curve in the x - y plane

w parameterisation $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j}$



$$\Delta l = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

Derivation

Since $\Delta l = \sqrt{(\Delta x)^2 + (\Delta y)^2}$

$$\frac{\Delta l}{\Delta t} = \frac{\sqrt{(\Delta x)^2 + (\Delta y)^2}}{\Delta t}$$

$$= \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2}$$

Let $\Delta t \rightarrow 0$, $\frac{\Delta l}{\Delta t}, \frac{\Delta x}{\Delta t}, \frac{\Delta y}{\Delta t} \rightarrow \frac{dl}{dt}, \frac{dx}{dt}, \frac{dy}{dt}$

$$\therefore \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

t is just a parameter.

Example

If $y = f(x)$, then $t = x$

$$\Rightarrow s = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Another form is gotten from using $\frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

$$s = \int_{t_1}^{t_2} \sqrt{\frac{d\vec{r}}{dt} \cdot \frac{d\vec{r}}{dt}} dt \rightarrow \left[\frac{dx}{dt} \right] \cdot \left[\frac{dx}{dt} \right] = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$$

By using the property $\vec{v} \cdot \vec{v} = |\vec{v}|^2$

$$\Rightarrow s = \int_{t_1}^{t_2} \left| \frac{d\vec{r}}{dt} \right| dt$$

This is just the little pieces of arc length, ds

Therefore, the 2 forms are

$$s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

or

$$s = \int_{t_1}^{t_2} \left| \frac{d\vec{r}}{dt} \right| dt$$

Arc Length, 3D

Since 1 dimension is added, $ds = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$

Using the same procedure as 2D, we get

$$S = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

or

$$S = \int_{t_1}^{t_2} \left| \frac{d\vec{r}}{dt} \right| dt$$

Line Integrals

Line integral is an integral used to accumulate a quantity along a curve

$$\int_C f ds$$

→ For each arc length piece ds , take the value of f there, then add all these little amount $f ds$

Arc length is the simplest line integral by setting $f = 1$, we are essentially just summing up all the little pieces of arc length ds $[S = \int_C 1 ds]$

Example 1 : [Line Integral in Scalar Field]

Consider the helix $\vec{r}(t) = \frac{3}{2} \cos(2\pi t) \hat{i} + \frac{3}{2} \sin(2\pi t) \hat{j} + t \hat{k}$, $0 \leq t \leq 3$.

If the density is $\rho(x, y, z) = x^2 + y^2 + z$, what is the mass?

Each tiny piece of wire, ds has tiny mass, dm

⇒

$$dm = \rho ds$$

$$m = \int_C \rho(x, y, z) ds$$

$$= \int_{t_1}^{t_2} \rho(\vec{r}(t)) \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt$$

$$= \int_0^3 \left\{ \left[\frac{3}{2} \cos(2\pi t) \right]^2 + \left[\frac{3}{2} \sin(2\pi t) \right]^2 + t \right\} \sqrt{[-3\sin(2\pi t)]^2 + [3\cos(2\pi t)]^2 + 1} dt$$

$$= \int_0^3 \left[\frac{9}{4} + t \right] \cdot \sqrt{9\pi^2 + 1} dt$$

$$= \sqrt{9\pi^2 + 1} \left[\frac{9}{4} t + \frac{t^2}{2} \right]_0^3$$

$$= \frac{45}{4} \sqrt{9\pi^2 + 1}$$

Example 2: [Line Integral in Scalar Field]

Evaluate the line integral of $f(x,y) = \frac{x^2+y^2}{4} + \frac{xy}{4}$ above circle of radius 2 at the origin.

Scalar field

To get that surface area, we just sum up the "height $\times ds$ " along the path

$$\int_C f(x,y) ds$$

Plot circle once

Curve, $C : \vec{r}(t) = 2\cos(t)\hat{i} + 2\sin(t)\hat{j}, 0 \leq t \leq 2\pi$

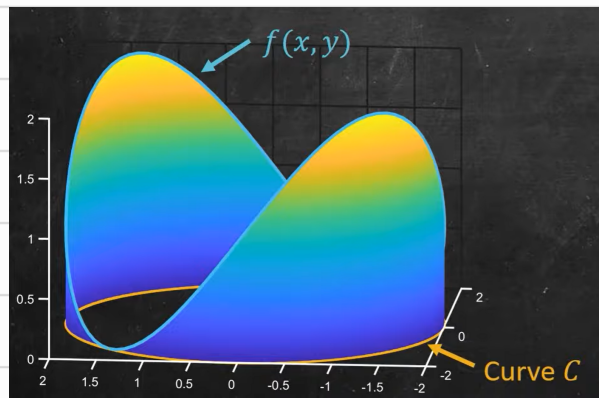
$$f(\vec{r}(t)) = \frac{(2\cos(t))^2 + (2\sin(t))^2}{4} + \frac{[2\cos(t)][2\sin(t)]}{4}$$

$$f(\vec{r}(t)) = 1 + 2\sin 2t$$

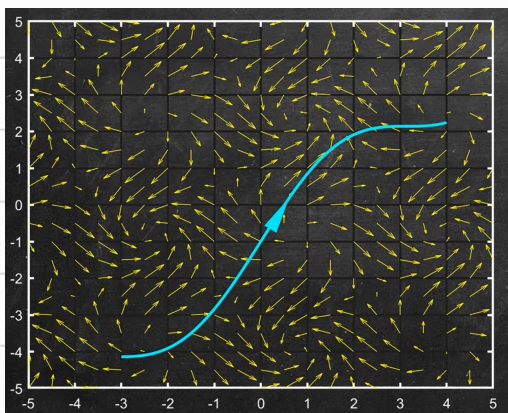
$$\therefore \int_0^{2\pi} [1 + 2\sin 2t] \times \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

$$= \int_0^{2\pi} [1 + 2\sin 2t] \times 2 dt$$

$$= 4\pi$$



Line integrals in Vector Fields



Consider a vector field $\vec{F}$ and a curve C

What is the work done by that field if a particle moves along the curve?

The contribution of the field along ds is (proportion of the force in direc $\vec{T}$) $\times ds$

$$\Rightarrow \vec{F} \cdot \vec{T} \times ds$$

$\therefore$ The total work done by $\vec{F}$ on the particle along the entire curve, C is

$$\int_C \vec{F} \cdot \vec{T} ds$$

$$\int_C \vec{F} \cdot \frac{d\vec{r}}{ds} ds$$

$$ds = \frac{ds}{dt} dt$$

$$\int_C \vec{F}(\vec{r}(t)) \cdot d\vec{r}$$

$$\int_{t_1}^{t_2} \vec{F}(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt} dt$$

Example 3: [Line integral in Vector Fields]

Find the work done by the force field

$$\mathbf{F} = \langle (x-y), (x+y), (z+2x) \rangle$$

in a moving particle along a straight line $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ to $\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$
position at $t=0$

$$\begin{aligned} \text{Curve, } C : \vec{r}(t) &= \vec{r}_0 + t \vec{v} \quad \begin{array}{l} \text{direc. vec} \\ \text{of line} \end{array} \\ &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \quad \begin{array}{l} \text{so } t=0, \vec{r}(t) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ t=1, \vec{r}(t) = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \end{array} \\ &= \begin{pmatrix} t \\ 2t \\ 5t \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{Work done} &= \int_{t_1}^{t_2} \left[\vec{F} \cdot \frac{d\vec{r}}{dt} \right] dt \\ &= \int_0^1 \left[\begin{pmatrix} x-y \\ x+y \\ z+2x \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \right] dt \\ &= \int_0^1 \left[\begin{pmatrix} t-2t \\ t+2t \\ 5t+2t \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \right] dt \\ &= \int_0^1 (-t + 6t + 35t) dt \\ &= 20 \text{ J} \end{aligned}$$

Conservative Vector Fields

Vector field whose line integral depends only on the start & end points, not on the path taken.

$$\vec{F} = \nabla f$$

This is the flow of thought

- 1) Assume we have a scalar function, f which maps each point in space to a particular value
- 2) Assume the vector field, $\vec{F}$ is gotten from the gradient of scalar function, f
[f is a scalar potential function to $\vec{F}$]
- 3) This means $\vec{F}$ have an underlying "map" that tells u each point's particular value, and $\vec{F}$ is merely just showing the gradient of it.
- 4) This means it doesn't matter how I reach point A, if I'm at point A, I have point A's value that is determined by the scalar potential function, f of $\vec{F}$.

$$\begin{aligned}\therefore \int_{t_1}^{t_2} \vec{F} \cdot \frac{d\vec{r}}{dt} dt &= \int_{t_1}^{t_2} \nabla f \cdot \frac{d\vec{r}}{dt} dt \\ &= f(t_2) - f(t_1)\end{aligned}$$

Other properties

$$\nabla \times \vec{F} = 0 \quad [\text{Curl} = 0]$$

$$\oint_C \vec{F} \cdot d\vec{r} = 0 \quad [\text{work done closed loop} = 0]$$

Trick for 2D

$$\begin{aligned}\vec{F} &= \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \\ \Rightarrow \vec{F} &= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \\ &= P \hat{i} + Q \hat{j}\end{aligned}$$

Since we know $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

$$\text{So } \boxed{\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}}$$

$\therefore$ If $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, the field is conservative

Example 4 : [conservative vec. field]

Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F} = \langle 2xz, 2yz, x^2 + y^2 \rangle$

and the curve, $C : \vec{r}(t) = \langle \cos(2\pi t), 2\sin(\frac{\pi}{2}t), (t^3 + \cos(2\pi t)) \rangle, 0 \leq t \leq 1$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 \vec{F}(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt} dt \quad \text{[This is way to complicated for the given } \vec{F} \text{ \& } C \text{]}$$

So we check if $\vec{F}$ is conservative.

$$\nabla \times \vec{F} = \begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} 2xz \\ 2yz \\ x^2 + y^2 \end{pmatrix}$$

$$= \begin{pmatrix} 2y - 2y \\ -(2x - 2x) \\ 0 - 0 \end{pmatrix}$$

$$= \vec{0} \Rightarrow \text{conservative}$$

$$\therefore \vec{F} = \nabla f$$
$$\begin{pmatrix} 2xz \\ 2yz \\ x^2 + y^2 \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix}$$

$$\Rightarrow \frac{\partial f}{\partial x} = 2xz ; \frac{\partial f}{\partial y} = 2yz ; \frac{\partial f}{\partial z} = x^2 + y^2$$

$$f = x^2z + C_1(y, z) \quad \text{--- ①}$$

↓

$$\frac{\partial f}{\partial y} = \frac{\partial C_1}{\partial y} \quad \text{--- ②}$$

Using ②,

$$\frac{\partial C_1}{\partial y} = 2yz$$

$$C_1(y, z) = y^2z + C_2(z)$$

$$\therefore f = x^2z + y^2z + C_2(z)$$

↓

$$\therefore \frac{\partial f}{\partial z} = x^2 + y^2 + \frac{dC_2}{dz}$$

$$x^2 + y^2 + \frac{dC_2}{dz} = x^2 + y^2$$

$$\frac{dC_2}{dz} = 0$$

$$C_2(z) = C_3$$

$$\therefore f(x, y, z) = x^2z + y^2z + C_3$$

$$= (x^2 + y^2)z + C_3$$

$$\text{When } t = 0, \vec{r}(t) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$t = 1, \vec{r}(t) = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = f(\vec{r}(1)) - f(\vec{r}(0))$$

$$= f(1, 2, 2) - f(1, 0, 1)$$

$$= [(1^2 + 2^2)2 + C_3] - [(1^2 + 0^2)1 + C_3]$$

$$= 9$$

